

Cyber Kill Chain & Attack Lifecycles

The Cyber Kill Chain (Lockheed Martin, 2011)

Modeled on military kill chains, it breaks an intrusion into seven phases:

Reconnaissance – research the target (OSINT, scanning)

Weaponization – craft the payload (malware + exploit + delivery)

Delivery – transmit the weapon (email, USB, web drive-by)

Exploitation – trigger the payload (buffer overflow, phishing click)

Installation – establish persistence (backdoor, service install)

Command & Control (C2) – open a channel to the attacker

Actions on Objectives – exfiltrate, encrypt, destroy

Recon -> Weaponize -> Deliver -> Exploit -> Install -> C2 -> Act

Defensive insight: You can disrupt the chain at ANY phase. Stop delivery → no infection. Stop C2 → no actions on objectives. Defenders who think in phases can pick the cheapest place to break the chain.

Offensive insight: The fewer phases the attacker must complete, the better their opsec. Phishing to clipboard (deliver+exploit in one click) is cheaper than a multi-stage C2 deployment.

MITRE ATT&CK

The modern standard. Rather than a linear chain, ATT&CK maps **tactics** (the "why" – columns) and **techniques** (the "how" – rows) from real-world observations.

15 enterprise tactics include: Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access, Discovery, Lateral Movement, Collection, Command and Control, Exfiltration, Impact.

Techniques like `T1133 External Remote Services`, `T1021.001 Remote Desktop Protocol` reference both tactic and procedure – enabling standardized detection and reporting.

Using the models together

Kill chain describes `one intrusion's arc`

ATT&CK describes `the techniques in detail`, mapped across the whole MITRE matrix

Defenders map detections to techniques (MITRE says "detect T1133, not 'attacker'")

Blue teams can organize detections per tactic playbook; red teams can plan procedure chains per tactic

MITRE ATT&CK references a technique by ID: e.g., T1572 "Protocol Tunneling" in the Command and Control tactic. Adversaries' documents and threat intel feeds publish these IDs so everyone speaks the same language.

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